

Data and weights
Matrix $\mathbf{Z}^{p1}$ of quantitative variables and weights $\mathbf{w}$

Weighted centering
$$\mathbf{Z}_{\mathbf{w}}^{p1} = \mathbf{D}_{\mathbf{w}}^{1/2} \cdot \mathbf{J}_{\mathbf{w}} \cdot \mathbf{Z}^{p1}$$

Trim outliers
Sort by ϕ , remove top $[\alpha \cdot n]$ observations
Obtain trimmed dataset $\mathbf{Z}_{\alpha}^{p1}$

Compute distances and proximity
Mahalanobis distances Δ_{mah}
Proximity scores $\phi(\mathbf{z}_i)$ for all i

Re-center trimmed data
Compute $\mathbf{Z}_{\mathbf{w},\alpha}^{p1} = \mathbf{D}_{\mathbf{w},\alpha}^{1/2} \cdot \mathbf{J}_{\mathbf{w},\alpha} \cdot \mathbf{Z}_{\alpha}^{p1}$

Estimate robust covariance
$$\hat{\mathbf{S}}_{\alpha} = \mathbf{Z}_{\alpha}^{p1\top} \cdot \mathbf{J}_{\mathbf{w},\alpha}^{\top} \cdot \mathbf{D}_{\mathbf{w},\alpha} \cdot \mathbf{J}_{\mathbf{w},\alpha} \cdot \mathbf{Z}_{\alpha}^{p1}$$

Compute robust Mahalanobis distance
$$\delta^2(\mathbf{z}_i, \mathbf{z}_j) = (\mathbf{z}_i - \mathbf{z}_j)^{\top} \hat{\mathbf{S}}_{\alpha}^{-1} (\mathbf{z}_i - \mathbf{z}_j)$$